

restructured by integrating StarNet and SlimNeck, and a Self-Attention Detect Head (SADH) is designed to enhance feature extraction. These modifications result in a lightweight network without compromising recognition accuracy. Finally, the network's performance is validated through ablation and comparative experiments. Results show that YOLO-S³ features a highly lightweight network design. When the signal-to-jamming ratio varies from -10 to 0 dB and the Signal-to-Noise Ratio (SNR) is ≥ 0 dB, the network achieves an impressive average recognition accuracy of 99.5%. Even when the SNR decreases to -10 dB, it maintains a robust average recognition accuracy of 95.5%, exhibiting strong performance under low SNR conditions. These findings provide a promising solution for the real-time recognition of composite jamming signals on resource-constrained platforms such as airborne radar signal processors and portable electronic devices.

Key words: Radar composite jamming; Lightweight; YOLO-S³; Low Signal-to-Noise Ratio (SNR); Jamming signal recognition

1 引言

随着电子信息与人工智能技术的深度融合,现代雷达体系正向智能化方向快速发展^[1]。作为雷达体系的核心节点,雷达智能抗干扰系统性能的优劣直接影响整个体系的可靠性与稳定性。该系统通过

一种基于卷积神经网络的干扰信号识别算法。该方法将干扰设备占用的通道信息转换为频谱图像,在 JNR 为 30 dB 条件下实现了约 90% 的识别准确率。Wang 等人^[9]采用短时傅里叶变换生成 5 种单一干扰信号的时频图像,并基于三维残差网络实现干扰信

restructured by integrating StarNet and SlimNeck, and a Self-Attention Detect Head (SADH) is designed to enhance feature extraction. These modifications result in a lightweight network without compromising recognition accuracy. Finally, the network's performance is validated through ablation and comparative experiments. Results show that YOLO-S1 features a highly lightweight network design. When the signal-tojamming ratio varies from -10 to 0 dB and the Signal-to-Noise Ratio (SNR) is 0 dB, the network achieves an impressive average recognition accuracy of 99.5%. Even when the SNR decreases to 10 dB, it maintains a robust average recognition accuracy of 95.5%, exhibiting strong performance under low SNR conditions. These findings provide a promising solution for the real-time recognition of composite jamming signals on resourceconstrained platforms such as airborne radar signal processors and portable electronic devices.

Key words: Radar composite jamming; Lightweight; YOLO-S3; Low Signal-to-Noise Ratio (SNR); Jamming signal recognition

1 引言

随着电子信息与人工智能技术的深度融合,现代雷达体系正向智能化方向快速发展^[1]。作为雷达体系的核心节点,雷达智能抗干扰系统性能的优劣直接影响整个体系的可靠性与稳定性。该系统通过干扰感知模块实现对干扰类型及参数的精准识别,并依托抗干扰决策模块生成最优应对策略^[23]。在此过程中,高效且轻量的干扰识别技术是确保资源受限硬件设备实现有效抗干扰的关键。因此,研究雷达干扰的智能感知方法具有重要的科研价值与工程意义。